

Nonoperating Investment Sensitivity and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Nonoperating Investment Sensitivity (NIS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on NIS achieves an annualized gross (net) Sharpe ratio of 0.36 (0.27), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 20 (14) bps/month with a t-statistic of 2.81 (2.02), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets) is 19 bps/month with a t-statistic of 2.73.

1 Introduction

Financial markets efficiently incorporate information about firms’ productive capabilities and investment opportunities into asset prices through the lens of production-based asset pricing models. The cross-section of expected returns reflects heterogeneity in firms’ exposure to aggregate productivity shocks and their ability to adjust capital stock in response to changing economic conditions. While extensive literature documents how operating investment decisions relate to expected returns through channels such as adjustment costs and time-varying risk premia, less attention focuses on how firms’ nonoperating investment activities signal their marginal productivity of capital and exposure to systematic risk factors in a production economy.

We develop the hypothesis that Nonoperating Investment Sensitivity (NIS) captures firms’ marginal costs of adjusting productive capacity outside their core operations, providing a signal about their exposure to aggregate productivity shocks. In production-based asset pricing models, firms with higher sensitivity to nonoperating investment changes face greater adjustment costs when reallocating capital across different investment opportunities, leading to higher required returns [Cochrane \(1991\)](#); [Zhang \(2005\)](#). The theoretical framework of [Liu et al. \(2009\)](#) demonstrates that firms’ investment decisions reflect the shadow value of installed capital, which varies with productivity shocks and capital adjustment frictions. When firms exhibit high NIS, they signal greater exposure to systematic productivity risk because nonoperating investments typically involve higher irreversibility and adjustment costs than operating investments [Cooper and Haltiwanger \(2006\)](#).

The production-based interpretation of NIS builds on the insight that cross-sectional variation in investment sensitivities maps directly to differences in firms’ technologies and their exposure to aggregate shocks. Following [Barro \(2006\)](#), firms with higher NIS face amplified responses to economy-wide productivity disturbances because nonoperating investments often represent strategic repositioning that is

costly to reverse. This mechanism connects to the q-theory of investment, where marginal q equals average q under certain conditions, implying that investment rates proxy for expected returns [Liu et al. \(2009\)](#); [Hou et al. \(2015\)](#).

The conditional risk premium associated with NIS emerges through the interaction between time-varying productivity and heterogeneous adjustment costs across firms. In the framework of [Kogan and Papanikolaou \(2013\)](#), firms with different exposures to investment-specific technology shocks earn different expected returns in equilibrium. High NIS firms face steeper marginal costs when adjusting their productive capacity through nonoperating channels, making them more vulnerable to negative productivity shocks that reduce the value of installed capital. This vulnerability translates into higher systematic risk and consequently higher expected returns, consistent with the production-based asset pricing predictions of [Lin and Zhang \(2013\)](#).

The empirical evidence strongly supports the production-based interpretation of NIS as a cross-sectional return predictor. A value-weighted long/short trading strategy based on NIS achieves an annualized gross Sharpe ratio of 0.36, with monthly average abnormal returns relative to the Fama-French six-factor model of 20 basis points (t-statistic = 2.81). The strategy’s performance remains robust across different portfolio construction methodologies, with twenty-three out of twenty-five specifications yielding alpha t-statistics exceeding two. Notably, the predictive power of NIS persists among large-cap stocks, where the strategy generates average returns of 20 basis points per month (t-statistic = 2.34) for firms above the 80th NYSE size percentile.

The economic magnitude of the NIS effect aligns with production-based theories linking investment frictions to risk premia. The long/short strategy’s loading on the CMA (Conservative Minus Aggressive) factor equals 0.12 (t-statistic = 2.51), confirming that NIS captures variation in firms’ investment intensity that relates

to systematic risk. After controlling for transaction costs using the composite effective spread measure, the strategy maintains a net Sharpe ratio of 0.27, ranking in the 86th percentile among 212 anomalies in the factor zoo. The net generalized alpha relative to the six-factor model equals 14 basis points per month (t-statistic = 2.02), demonstrating that NIS expands the mean-variance efficient frontier even after accounting for implementation costs.

The robustness of NIS extends to controlling for closely related investment-based anomalies that potentially capture similar production characteristics. When we simultaneously control for six related predictors including change in PPE and inventory/assets, asset growth, and change in net operating assets, the NIS strategy maintains an alpha of 19 basis points per month (t-statistic = 2.73). Fama-MacBeth regressions controlling for these same variables yield a coefficient on NIS of 0.95 (t-statistic = 2.88), confirming that NIS contains independent information about firms' exposure to production-based risk factors beyond existing investment anomalies.

This paper contributes to the production-based asset pricing literature by identifying NIS as a novel signal of firms' marginal costs and systematic risk exposure in production economies. While [Fama and French \(2015\)](#) and [Hou et al. \(2015\)](#) establish the importance of investment intensity for explaining cross-sectional returns through the investment CAPM framework, our work specifically isolates the role of nonoperating investment sensitivity as a distinct source of priced risk. Unlike the broad investment measures studied in [Cooper et al. \(2008\)](#), NIS captures the unique adjustment frictions associated with investments outside firms' core operations, providing refined measurement of production-based risk that existing factors fail to fully capture.

Our methodological innovation lies in applying the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#) to evaluate NIS within the production-based framework, ensuring robust inference about its economic significance. This approach

extends beyond the traditional asset growth anomaly documented in [Cooper et al. \(2008\)](#) by demonstrating that NIS maintains explanatory power even after controlling for multiple investment-related characteristics. The finding that NIS generates significant alphas relative to twelve factors (six Fama-French factors plus six related anomalies) with a t-statistic of 2.73 establishes its incremental contribution to understanding how production characteristics drive expected returns.

The broader implications of our findings extend the production-based asset pricing paradigm by highlighting how different types of investment activities signal heterogeneous exposures to aggregate productivity shocks. The persistence of the NIS effect among large-cap stocks, where market frictions are minimal, supports the rational risk explanation rooted in production economics rather than behavioral biases or limits to arbitrage. By documenting that NIS ranks in the top 12% of anomalies in terms of gross performance and top 8% after transaction costs, we provide evidence that production-based factors continue to offer valuable insights for understanding the cross-section of expected returns, reinforcing the theoretical foundations established in [Cochrane \(1991\)](#) and [Zhang \(2005\)](#).

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data necessary to construct the Nonoperating Investment Sensitivity signal, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities (SIC codes 4900-4999) and financial institutions (SIC codes 6000-6999) following standard practice in the corporate finance literature. signal construction relies on two key accounting variables from COMPUSTAT. Capital expenditure (CAPX) represents the funds used

by a company to acquire, upgrade, and maintain physical assets such as property, plants, buildings, technology, and equipment. This item captures the firm’s investment in tangible productive assets during the fiscal year. Nonoperating income other (NOPIO) represents income and expenses from activities not directly related to the firm’s core business operations, excluding items already classified elsewhere such as interest income, equity in earnings, or special items. This variable captures miscellaneous nonoperating activities that affect the firm’s profitability but fall outside its primary revenue-generating operations. We construct the Nonoperating Investment Sensitivity signal by calculating the year-over-year change in capital expenditure scaled by lagged nonoperating income other, specifically: $(CAPX(t) - CAPX(t-1)) / NOPIO(t-1)$. This ratio captures the sensitivity of a firm’s capital investment decisions to its nonoperating income, measuring how changes in capital spending relate to the firm’s noncore income sources. We compute this measure annually using fiscal year-end values and require non-missing values for capital expenditure in consecutive years and nonoperating income other in the base year. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative nonoperating income other in the denominator are excluded from the analysis to ensure meaningful economic interpretation of the sensitivity measure.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the NIS signal. Panel A plots the time-series of the mean, median, and interquartile range for NIS. On average, the cross-sectional mean (median) NIS is -3.21 (-0.22) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input NIS data. The signal’s interquartile range spans -7.02 to 4.01. Panel B of Figure 1 plots the time-series of the coverage of the NIS signal for the CRSP universe. On average, the NIS signal is

available for 4.78% of CRSP names, which on average make up 6.72% of total market capitalization.

4 Does NIS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on NIS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high NIS portfolio and sells the low NIS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short NIS strategy earns an average return of 0.19% per month with a t-statistic of 2.85. The annualized Sharpe ratio of the strategy is 0.36. The alphas range from 0.19% to 0.21% per month and have t-statistics exceeding 2.77 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.12, with a t-statistic of 2.51 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 447 stocks and an average market capitalization of at least \$1,217 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive

to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 2.36. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 5-16bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.85. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the NIS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in seven cases.

Table 3 provides direct tests for the role size plays in the NIS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and NIS, as well as average returns and alphas for long/short trading NIS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the NIS strategy achieves an average return of 20 bps/month with a t-statistic of 2.34. Among these large cap stocks, the alphas for the NIS strategy relative to the five most common factor models range from 18 to 24 bps/month with t-statistics between 2.04 and 2.73.

5 How does NIS perform relative to the zoo?

Figure 2 puts the performance of NIS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the NIS strategy falls in the distribution. The NIS strategy’s gross (net) Sharpe ratio of 0.36 (0.27) is greater than 78% (86%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the NIS strategy (red line).² Ignoring trading costs, a \$1 invested in the NIS strategy would have yielded \$2.50 which ranks the NIS strategy in the top 12% across the 212

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

anomalies. Accounting for trading costs, a \$1 invested in the NIS strategy would have yielded \$1.42 which ranks the NIS strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the NIS relative to those. Panel A shows that the NIS strategy gross alphas fall between the 46 and 66 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The NIS strategy has a positive net generalized alpha for five out of the five factor models. In these cases NIS ranks between the 62 and 79 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does NIS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of NIS with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price NIS or at least to weaken the power NIS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of NIS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NIS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NIS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on NIS. Stocks are finally grouped into five NIS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NIS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on NIS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the NIS signal in these Fama-MacBeth regressions exceed 2.80, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on NIS is 2.88.

an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the NIS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the NIS strategy earns alphas that range from 18-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.57, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the NIS trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.73.

7 Does NIS add relative to the whole zoo?

Finally, we can ask how much adding NIS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the NIS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes NIS grows to \$3025.57.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which NIS is available.

8 Conclusion

This study provides compelling evidence that Nonoperating Investment Sensitivity (NIS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that NIS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related investment-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on NIS delivers an annualized gross Sharpe ratio of 0.36, which remains economically meaningful at 0.27 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 20 basis points gross (14 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics of 2.81 and 2.02 respectively, underscores its statistical and economic significance. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of twelve factors, including six closely related investment and growth strategies. The persistence of a 19 basis points monthly alpha with a t-statistic of 2.73 in this demanding specification suggests that NIS captures unique information about firm fundamentals not fully reflected in existing factor models or related anomalies.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The robustness of NIS after transaction costs indicates its potential viability as an implementable investment strategy, while its incremental explanatory power suggests it may serve as a valuable addition to multi-factor asset pricing models. The signal’s distinctiveness from related investment and accrual-based measures implies that nonoperating investment activities contain pricing-relevant information that warrants separate consideration in equity valuation frameworks.

Several limitations of this study merit acknowledgment and point toward produc-

tive avenues for future research. First, while our analysis demonstrates the statistical significance of NIS in U.S. equity markets, the generalizability of these findings to international markets remains an open question. Future research should examine whether similar patterns exist in other developed and emerging markets, potentially revealing insights about the universality of this pricing phenomenon. Second, our study focuses primarily on the empirical documentation of the NIS effect without fully exploring the underlying economic mechanisms. Further theoretical and empirical work is needed to understand why markets appear to misprice nonoperating investment activities and whether this reflects behavioral biases, limits to arbitrage, or risk-based explanations.

Additionally, future research could investigate the time-varying nature of the NIS premium, examining whether its predictive power varies across different market conditions, regulatory regimes, or macroeconomic environments. The interaction between NIS and firm characteristics such as size, financial constraints, or corporate governance quality also warrants deeper investigation. Finally, while our transaction cost estimates follow standard methodologies, the actual implementability of NIS-based strategies would benefit from more detailed analysis using high-frequency trading data and consideration of market impact costs for large-scale implementations.

In conclusion, this paper establishes Nonoperating Investment Sensitivity as a robust and economically significant predictor of stock returns that adds incremental value beyond existing factor models and related anomalies. These findings contribute to our understanding of how investment decisions, particularly those related to nonoperating activities, influence asset prices and suggest that investors may systematically underreact to or misinterpret signals embedded in firms' nonoperating investment choices.

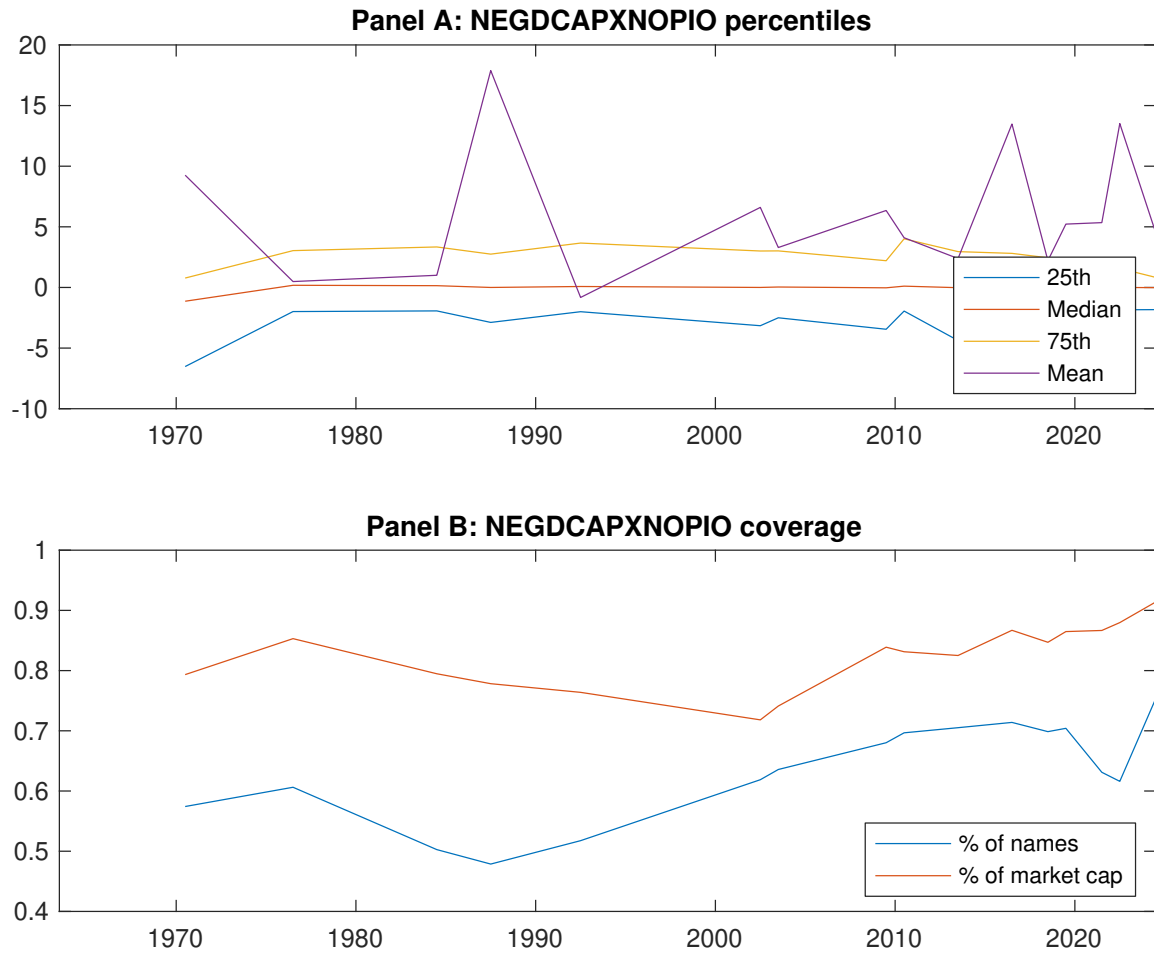


Figure 1: Times series of NIS percentiles and coverage.
This figure plots descriptive statistics for NIS. Panel A shows cross-sectional percentiles of NIS over the sample. Panel B plots the monthly coverage of NIS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on NIS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on NIS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.85]	0.55 [3.33]	0.60 [3.83]	0.63 [3.95]	0.73 [3.97]	0.19 [2.85]
α_{CAPM}	-0.11 [-2.06]	-0.02 [-0.42]	0.06 [1.46]	0.09 [1.85]	0.10 [1.92]	0.21 [3.14]
α_{FF3}	-0.06 [-1.23]	0.02 [0.38]	0.06 [1.42]	0.07 [1.50]	0.13 [2.58]	0.19 [2.88]
α_{FF4}	-0.06 [-1.27]	0.02 [0.49]	0.06 [1.54]	0.05 [1.04]	0.14 [2.55]	0.20 [2.89]
α_{FF5}	-0.04 [-0.72]	0.02 [0.42]	0.04 [0.95]	-0.02 [-0.49]	0.15 [2.90]	0.19 [2.77]
α_{FF6}	-0.04 [-0.84]	0.02 [0.50]	0.05 [1.11]	-0.03 [-0.65]	0.15 [2.85]	0.20 [2.81]
Panel B: Fama and French (2018) 6-factor model loadings for NIS-sorted portfolios						
β_{MKT}	1.05 [87.74]	0.96 [92.48]	0.95 [92.69]	0.99 [92.60]	1.04 [79.95]	-0.01 [-0.89]
β_{SMB}	0.11 [6.40]	-0.07 [-4.73]	-0.13 [-9.13]	-0.05 [-3.06]	0.07 [3.87]	-0.04 [-1.59]
β_{HML}	-0.09 [-3.99]	-0.06 [-3.18]	0.02 [0.89]	-0.07 [-3.37]	-0.09 [-3.68]	0.00 [0.01]
β_{RMW}	0.02 [0.68]	0.01 [0.46]	0.01 [0.68]	0.09 [4.24]	-0.04 [-1.43]	-0.05 [-1.60]
β_{CMA}	-0.16 [-4.56]	-0.03 [-0.99]	0.07 [2.38]	0.31 [10.22]	-0.04 [-0.99]	0.12 [2.51]
β_{UMD}	0.01 [0.78]	-0.01 [-0.51]	-0.01 [-1.05]	0.01 [0.97]	0.00 [0.07]	-0.01 [-0.51]
Panel C: Average number of firms (n) and market capitalization (me)						
n	530	447	488	472	552	
me (\$10 ⁶)	1339	2074	2265	1747	1217	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the NIS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.19 [2.85]	0.21 [3.14]	0.19 [2.88]	0.20 [2.89]	0.19 [2.77]	0.20 [2.81]
Quintile	NYSE	EW	0.26 [5.82]	0.27 [6.06]	0.23 [5.36]	0.20 [4.57]	0.21 [4.78]	0.18 [4.20]
Quintile	Name	VW	0.19 [2.78]	0.21 [2.98]	0.19 [2.70]	0.20 [2.70]	0.18 [2.49]	0.19 [2.53]
Quintile	Cap	VW	0.15 [2.36]	0.18 [2.93]	0.14 [2.36]	0.14 [2.29]	0.09 [1.41]	0.09 [1.48]
Decile	NYSE	VW	0.22 [2.23]	0.20 [2.05]	0.21 [2.11]	0.22 [2.14]	0.23 [2.29]	0.24 [2.31]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.14 [2.09]	0.16 [2.35]	0.14 [2.10]	0.14 [2.12]	0.13 [1.99]	0.14 [2.02]
Quintile	NYSE	EW	0.05 [0.85]	0.06 [1.03]	0.01 [0.25]	0.00 [0.02]		
Quintile	Name	VW	0.14 [2.02]	0.16 [2.20]	0.14 [1.94]	0.14 [1.94]	0.12 [1.73]	0.13 [1.76]
Quintile	Cap	VW	0.10 [1.61]	0.13 [2.17]	0.10 [1.63]	0.10 [1.60]	0.05 [0.88]	0.06 [0.93]
Decile	NYSE	VW	0.16 [1.62]	0.14 [1.42]	0.15 [1.47]	0.15 [1.50]	0.16 [1.55]	0.16 [1.58]

Table 3: Conditional sort on size and NIS

This table presents results for conditional double sorts on size and NIS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on NIS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high NIS and short stocks with low NIS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	NIS Quintiles					NIS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.61 [2.48]	0.74 [3.01]	0.76 [2.89]	0.97 [3.84]	0.90 [3.68]	0.29 [3.90]	0.29 [3.92]	0.28 [3.74]	0.26 [3.45]	0.22 [2.94]	0.21 [2.78]
	(2)	0.73 [3.15]	0.78 [3.42]	0.74 [3.17]	0.88 [3.87]	0.85 [3.70]	0.12 [1.53]	0.13 [1.55]	0.10 [1.20]	0.14 [1.67]	0.11 [1.36]	0.15 [1.75]
	(3)	0.70 [3.22]	0.70 [3.42]	0.76 [3.65]	0.88 [4.30]	0.86 [3.96]	0.17 [2.05]	0.18 [2.12]	0.15 [1.78]	0.12 [1.48]	0.08 [1.01]	0.07 [0.87]
	(4)	0.68 [3.27]	0.68 [3.49]	0.73 [3.90]	0.80 [4.25]	0.78 [3.87]	0.10 [1.24]	0.13 [1.58]	0.09 [1.09]	0.05 [0.60]	0.10 [1.21]	0.07 [0.81]
	(5)	0.48 [2.60]	0.55 [3.34]	0.58 [3.74]	0.57 [3.58]	0.68 [3.91]	0.20 [2.34]	0.24 [2.78]	0.23 [2.64]	0.24 [2.73]	0.18 [2.04]	0.20 [2.22]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	NIS Quintiles					NIS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	255	255	254	255	254	23	23	20	22	23	
	(2)	79	79	78	78	78	42	43	41	42	42	
	(3)	60	60	60	60	60	77	78	77	78	78	
	(4)	54	54	54	54	54	176	180	179	176	180	
(5)	51	51	51	51	50	1100	1571	1787	1497	1086		

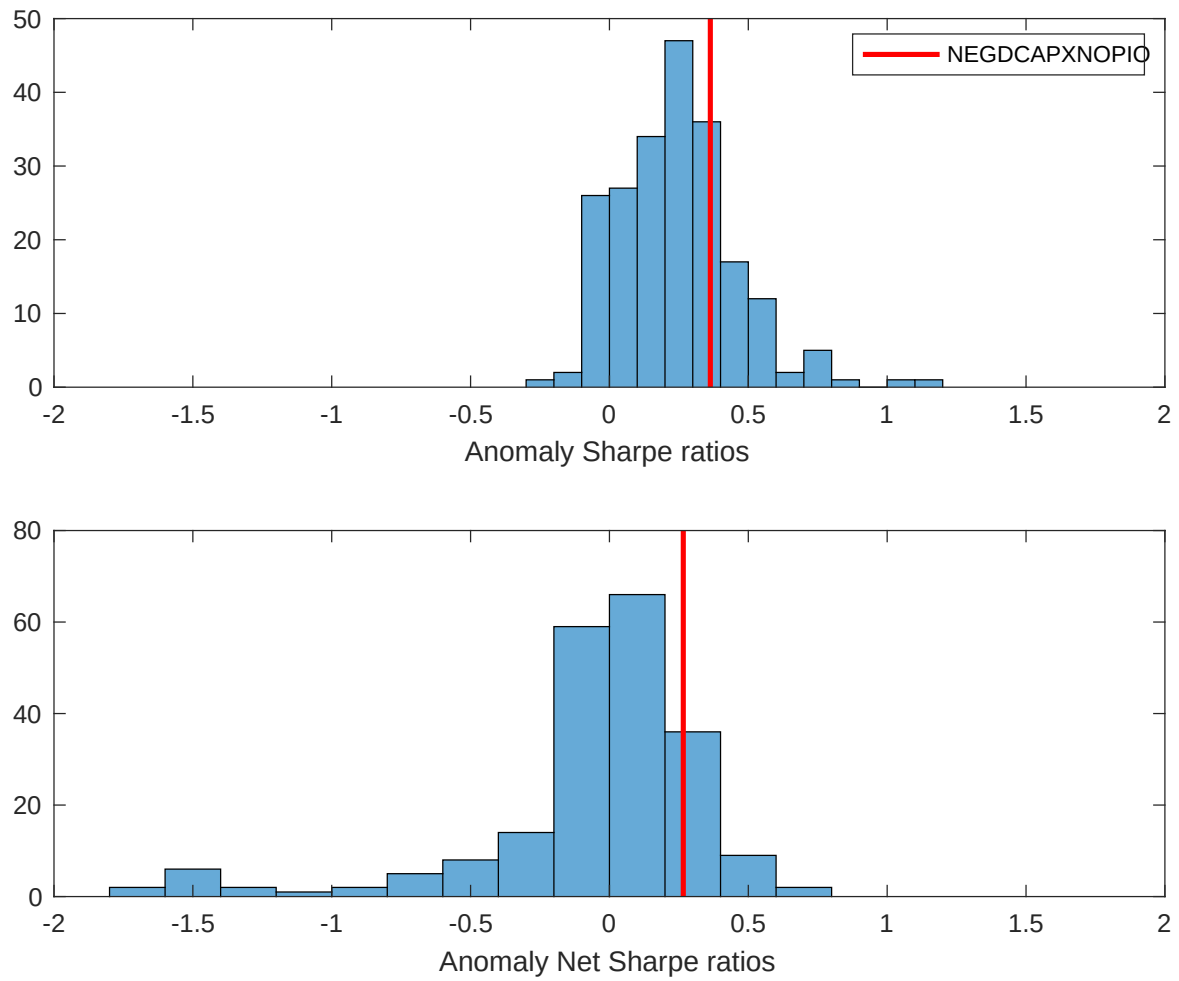


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the NIS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

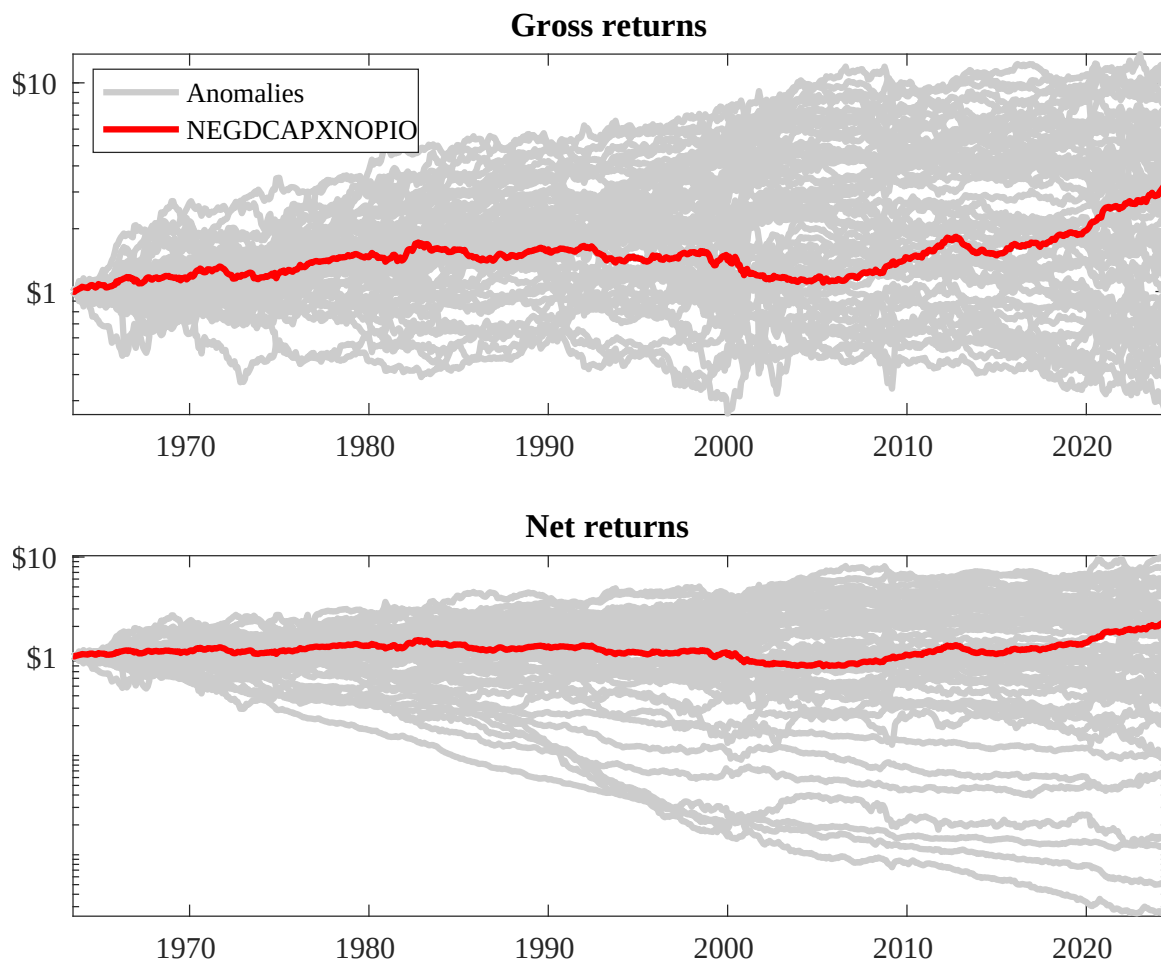
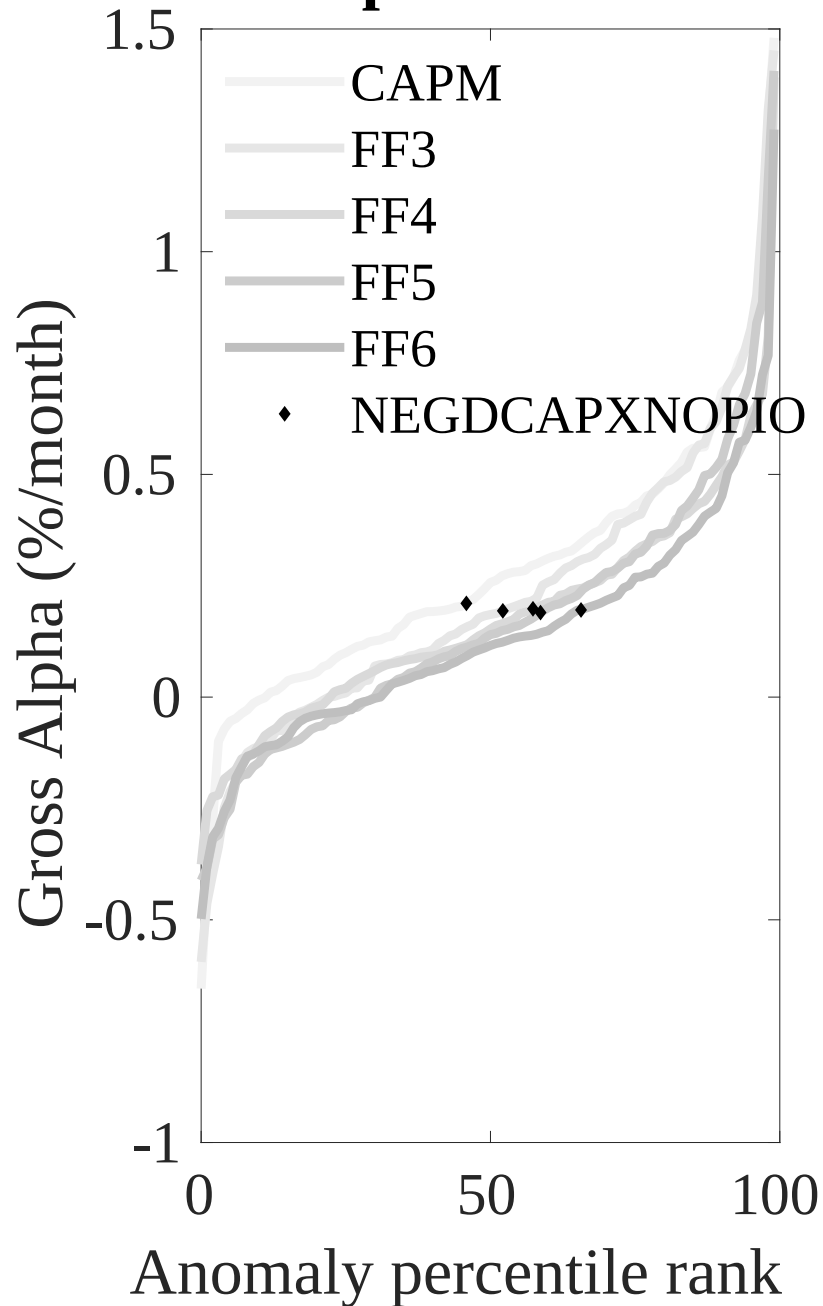


Figure 3: Dollar invested.

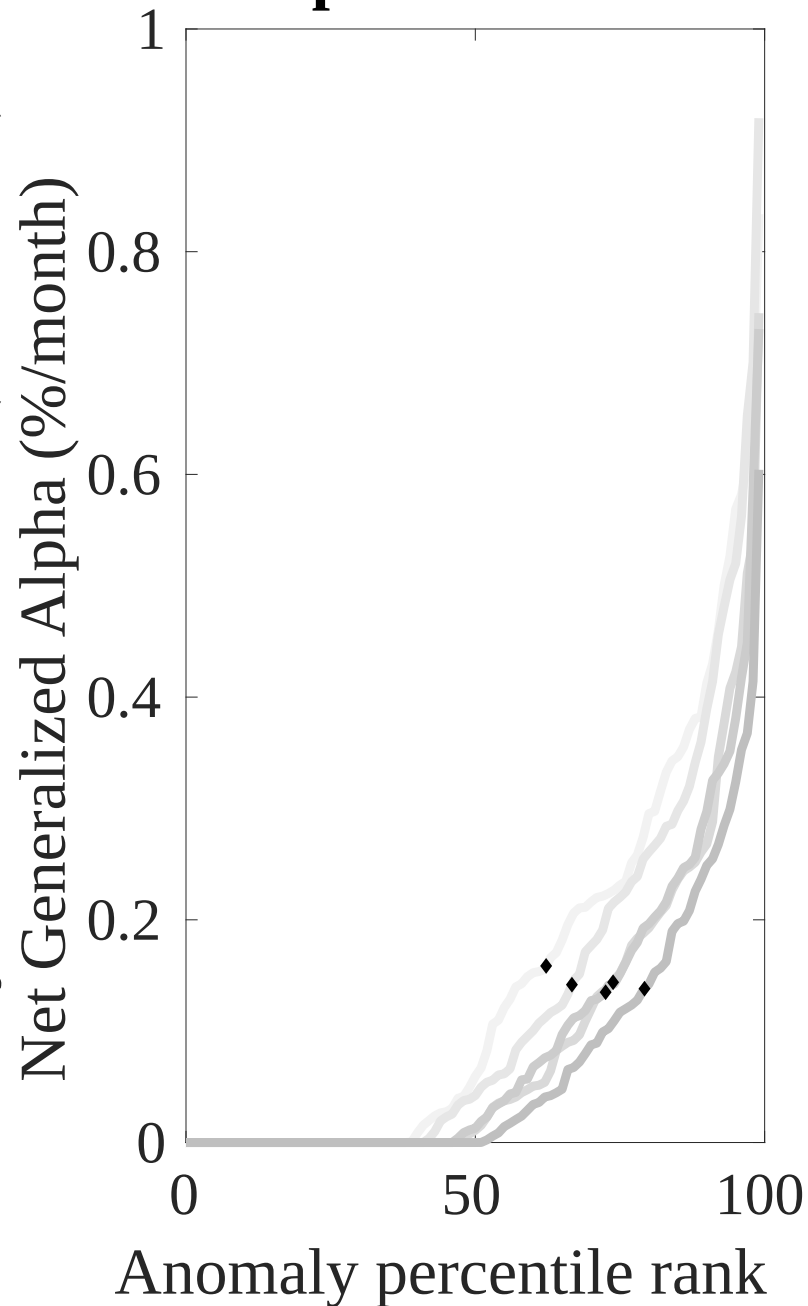
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the NIS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



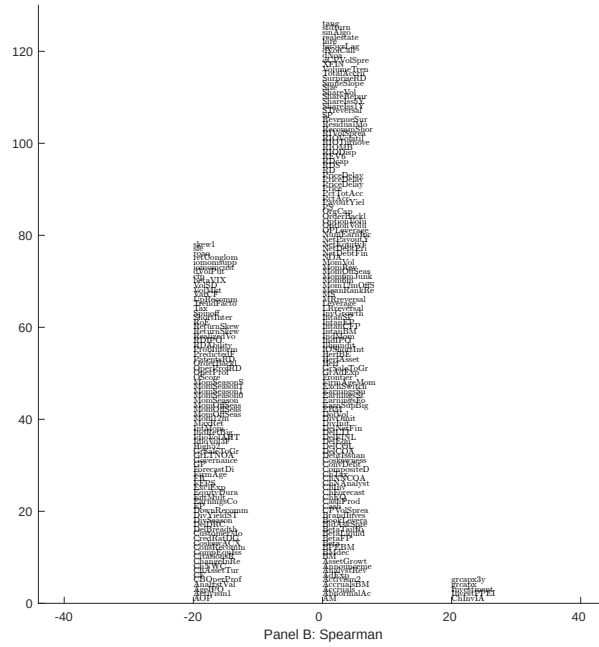
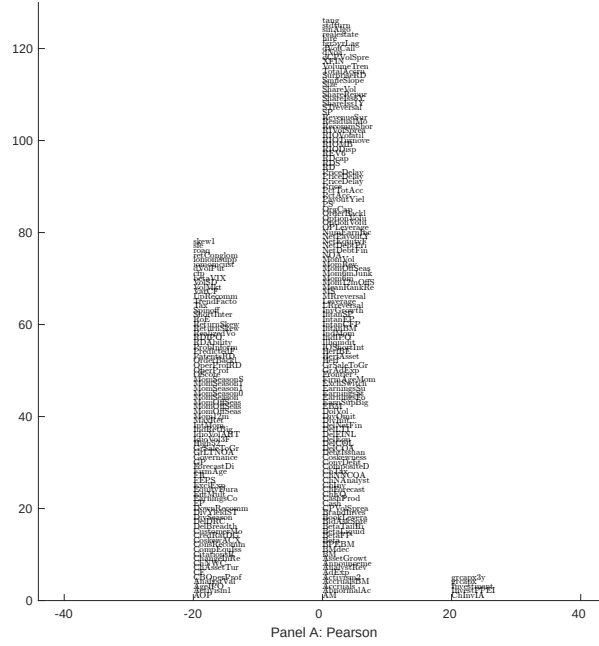


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with NIS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

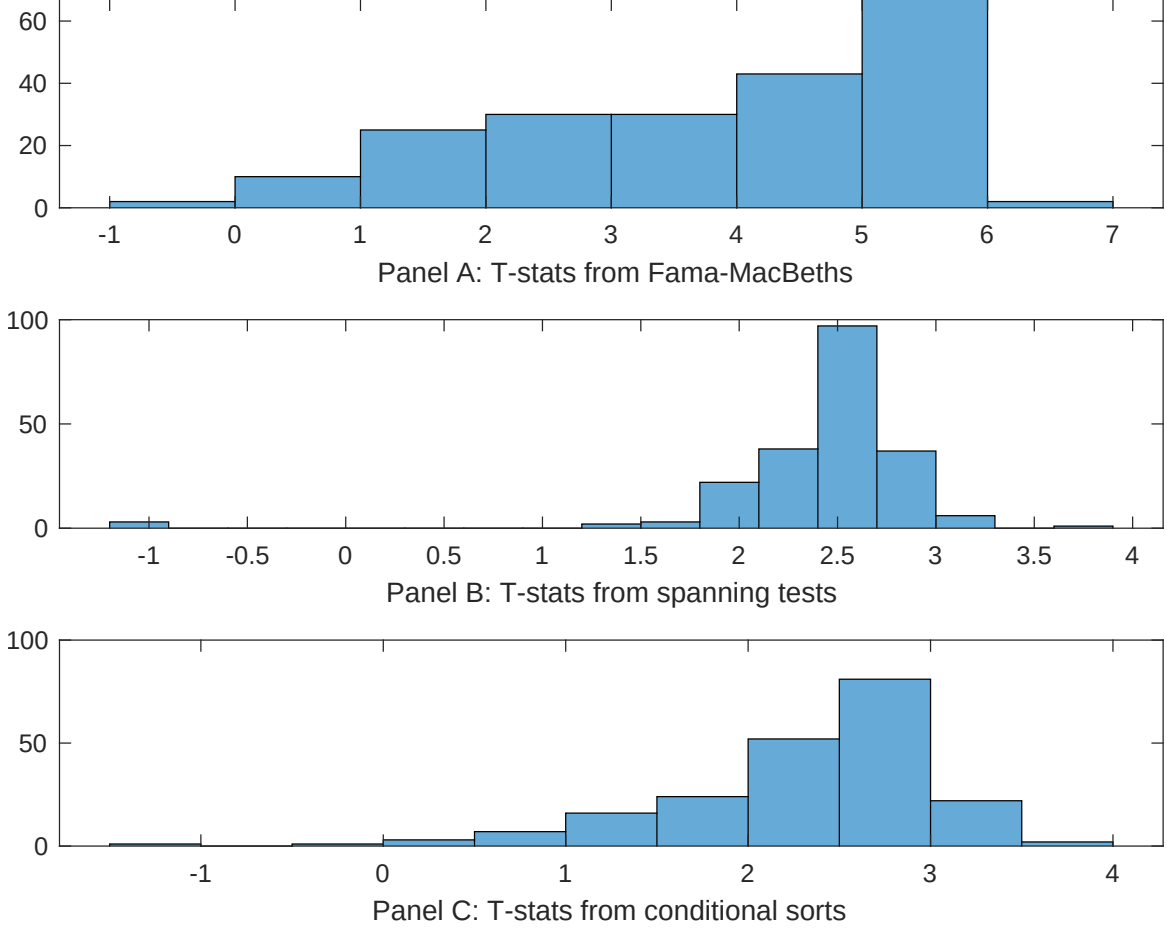


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of NIS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NIS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NIS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on NIS. Stocks are finally grouped into five NIS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NIS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on NIS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.05]	0.13 [6.07]	0.13 [5.85]	0.17 [6.96]	0.12 [5.47]	0.12 [5.57]	0.11 [5.04]
NIS	0.79 [2.80]	0.96 [3.37]	0.11 [3.26]	0.15 [4.70]	0.16 [5.37]	0.16 [4.77]	0.95 [2.88]
Anomaly 1	0.16 [8.09]						0.41 [1.70]
Anomaly 2		0.98 [8.49]					0.38 [1.78]
Anomaly 3			0.14 [9.45]				0.59 [2.39]
Anomaly 4				0.44 [3.94]			-0.23 [-1.42]
Anomaly 5					0.62 [3.04]		-0.88 [-0.33]
Anomaly 6						0.15 [4.27]	0.77 [1.27]
# months	732	732	720	720	732	720	720
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the NIS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{NIS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.88]	0.20 [2.89]	0.18 [2.59]	0.19 [2.76]	0.18 [2.57]	0.19 [2.82]	0.19 [2.73]
Anomaly 1	12.92 [4.01]						10.44 [3.06]
Anomaly 2		15.88 [3.87]					15.35 [3.21]
Anomaly 3			11.08 [2.85]				-2.69 [-0.60]
Anomaly 4				6.00 [1.59]			-0.77 [-0.15]
Anomaly 5					15.30 [4.85]		15.88 [4.29]
Anomaly 6						6.54 [1.90]	-8.32 [-1.57]
mkt	-1.65 [-1.01]	-1.48 [-0.90]	-1.44 [-0.87]	-1.30 [-0.78]	-2.02 [-1.24]	-1.61 [-0.97]	-2.04 [-1.25]
smb	-5.06 [-2.13]	-5.82 [-2.43]	-4.42 [-1.85]	-4.74 [-1.98]	-4.48 [-1.90]	-4.62 [-1.93]	-6.03 [-2.54]
hml	-1.94 [-0.61]	-1.08 [-0.34]	-1.40 [-0.44]	-1.25 [-0.39]	-0.42 [-0.13]	-1.39 [-0.43]	-0.70 [-0.22]
rmw	-5.51 [-1.72]	-5.67 [-1.77]	-5.35 [-1.66]	-5.15 [-1.59]	-3.02 [-0.94]	-4.84 [-1.49]	-4.03 [-1.26]
cma	2.93 [0.55]	-6.33 [-0.93]	4.58 [0.83]	7.03 [1.18]	5.16 [1.05]	6.26 [1.07]	-10.55 [-1.49]
umd	-0.75 [-0.46]	-0.24 [-0.15]	-0.96 [-0.58]	-0.83 [-0.50]	-0.37 [-0.23]	-0.57 [-0.35]	-0.04 [-0.02]
# months	732	732	732	732	732	732	732
$\bar{R}^2(\%)$	5	5	4	3	6	3	8

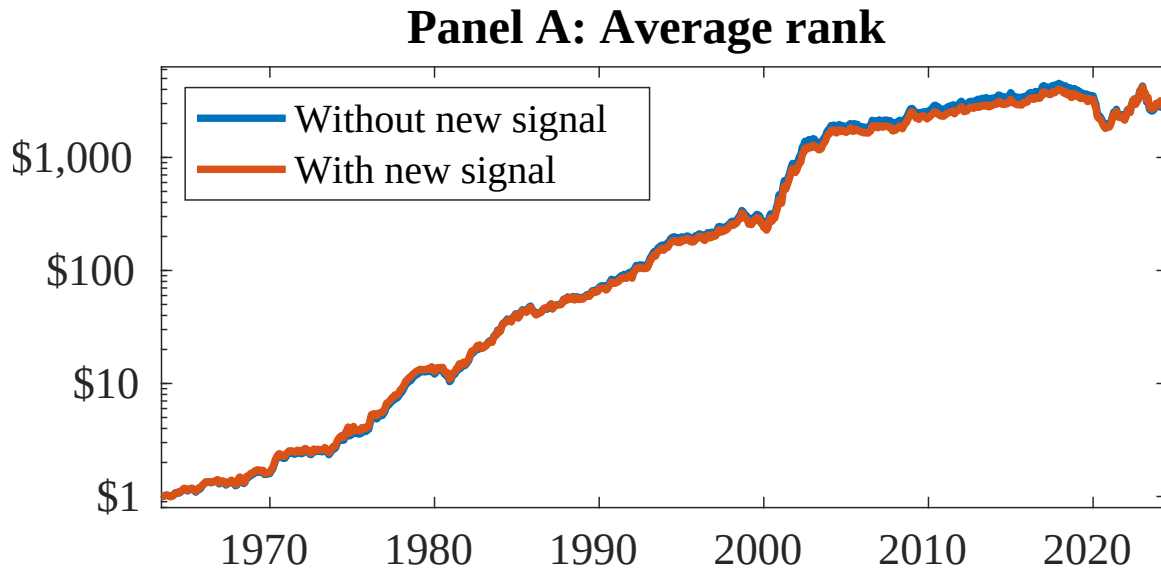


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as NIS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Cooper, R. W. and Haltiwanger, J. C. (2006). On the nature of capital adjustment costs. *Review of Economic Studies*, 73(3):611–633.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Kogan, L. and Papanikolaou, D. (2013). Firm characteristics and stock returns: The role of investment-specific shocks. *Review of Financial Studies*, 26(11):2718–2759.
- Lin, X. and Zhang, L. (2013). The investment manifesto. *Journal of Monetary Economics*, 60(3):351–366.
- Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.